Dr. Pradipta K. Banerjee

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DoB. 21.12.1977
Nationality: Indian

Education

2015 **Post Doctoral Fellow, Tsinghua University, Beijing**

Title: Facial Landmark Tracking for Analysis of Human Neuro-psychological Factors
Supervisor : Prof. Yu-Jin ZHANG, Department of Electronic Engineering

Tenure : 2 years (not completed)

2014 **PhD , University of Calcutta**

Title: Some Studies on Performance Improvement of Frequency Domain Correlation Technique for Face Recognition
Advisor: Prof Asit.K.Datta

2002-2004 **Master of Technology(Electrical Engineering), University of Calcutta**

Title:Multi-resolution Analysis in Block Pulse Function Domain using C.
Advisor:Prof. A.Deb, Prof. G. Sarkar
79.5%, 1st class

1999-2002 **Bachelor of Technology (Electrical Engineering), University of Calcutta**

Title:PC Based Transfer Function Identification using Block Pulse Function
Advisor:Prof. A.Deb, Prof. G. Sarkar
75.2%, 1st class

1996-1999 **Bachelors of Science (Physics Hons.), Vidyasagar College, University of Calcutta**, *62.38%, 1st class*

1994-1996 **Higer Secondary(WBCHSE), Barrackpore Govt. High School**, *77.8%, 1st division*

1994 **Secondary(WBBSE), Sagore Dutta High School**, *84.7%, 1st division*

Publications

Book

- Asit K. Datta, Madhura Datta, **Pradipta K. Banerjee**, Face Detection and Recognition : Theory and Practices, CRC Press, Taylor and Francis, 2015.

Book Chapters

1. **Pradipta K. Banerjee** *Phase-Encoded Cascaded Autoencoders based Correlation Filters and Adaptive Thresholding for Illumination Invariant Face Recognition*, Pattern Recognition(LNCS Series), Springer, 2024

2. Tiash Ghosh, **Pradipta K. Banerjee**, *Linear Regression Correlation Filter: An Application to Face Recognition*, Proceedings of 3rd International Conference on Computer Vision and Image Processing, Advances in Intelligent Systems and Computing, Springer, 2019

3. Tiash Ghosh, **Pradipta K. Banerjee**, A *Class Specific Representation Learning for Illumination Tolerant Face Recognition*, 2018, CCIS, Springer, 2018

4. Papia Banerjee, **Pradipta K. Banerjee**, Asit K. Datta, *Efficient Image Plane Rotation Invariant Frequency Domain Face Recognition Technique Using Eye Localization*, *Springer-Verlag Berlin Heidelberg*,2013

5. **Pradipta K. Banerjee**, Asit K. Datta, *Techniques of Frequency Domain Correlation for Face Recognition and its Photonic Implementation*, Ed.: A. Quaglia and C. M. Epifano, pp. 165-186 , Face Recognition: Methods,Applications and Technology, *Series: Computer Science, Technology and Applications, NOVA Publishers*, 2012

6. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Class Dependent 2D Correlation Filter for Illumination Tolerant Face Recognition*, M.K. Kundu et al. (Eds.):*Lecture Notes in Computer Science, Springer-Verlag Berlin Heidelberg,7143, 338-345*, 2012

SCI/Scopus Journals

1. A. Halder, S. Panja, S. Bhattacharya, D. Chakraborty, **Pradipta K. Banerjee**, *Deep Facial Feature Fusion and Voting Strategies for Enhanced Emotion Recognition*, *Procedia Computer Science, Elsevier* May,2025 **(Scopus)**

2. **Pradipta K. Banerjee**, Asit K. Datta, *Bandpass correlation filter for illumination and noise tolerant face recognition*, *Signal Image Video Processing, Springer,11,1, 9-16*, January 2017, **(SCI)**

3. **Pradipta K. Banerjee**, Asit K. Datta, *Class specic subspace dependent nonlinear correlation filtering for illumination tolerant face recognition*, *Pattern Recognition Letters, Elsevier, 36, 177-185*, 15 January 2014, **(SCI)**

4. **Pradipta K. Banerjee**, Asit K. Datta, *A preferential digital-optical correlator optimized by Particle Swarm technique for multi class face recognition*, *JOLT, Elsevier, 50, 33-42*, 2013, **(SCI)**

5. **Pradipta K. Banerjee**, Asit K. Datta, *Generalized Regression Neural Network Trained Preprocessing of Frequency Domain Correlation Filter for Improved Face Recognition and its Optical Implementation*, *JOLT, Elsevier, 45, 217-227*, 2013, **(SCI)**

6. **Pradipta K. Banerjee**, Asit K. Datta, *An Illumination Tolerant Class Specic 2D Subspace Based Face Recognition Technique Using Optimum Correlation Filter*, *Optik - International Journal for Light and Electron Optics, Elsevier, 124, Issue 17, 3173-3179*, September 2013,**(SCI)**

7. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *A frequency domain face recognition technique based on correlation plane features as input to a regression neural network*, *Procedia Computer Science, Elsevier, 2,75-82*,2010,**(Scopus)**

8. Jayanta K. Chandra,**Pradipta K. Banerjee**, Asit K. Datta, *Neural network trained morphological processing for the detection of defects in woven fabric*, *Journal of Textile Institute, Taylor Francis, 101, 699-706*, 2010,**(SCI)**

9. **Pradipta K. Banerjee**, Asit K. Datta, *Illumination and noise tolerant face recognition based on eigen-phase correlation lter modified by Mexican hat wavelet*, *Journal of Optics, Springer, 38,160-168*, 2008,**(SCI)**

International / National Conferences

1. **Pradipta K. Banerjee**, *Phase-Encoded Cascaded Autoencoders based Correlation Filters and Adaptive Thresholding for Illumination Invariant Face Recognition.* In: Antonacopoulos, A. et.al. (eds) Pattern Recognition. ICPR 2024. Lecture Notes in Computer Science, vol 15324. Springer, Cham. https://doi.org/10.1007/978-3-031-78383-8_12, 2025

2. A. Halder, S. Panja, S. Bhattacharya, D. Chakraborty, **Pradipta K. Banerjee**, *Deep Facial Feature Fusion and Voting Strategies for Enhanced Emotion Recognition*, (Provisional Acceptance), RTIP2R, 2024

3. D. Chakraborty, **Pradipta K. Banerjee**, *Facial emotion recognition using unconstrained minimum average correlation energy filter*, AIP Conference Proceedings , 2915(1) ,2024

4. D. Chakraborty, U. Sur, **Pradipta K. Banerjee,** *Random forest based fault classification technique for active power system networks.* IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE), 2020

5. T. Ghosh, **Pradipta K. Banerjee**, *Linear Regression Correlation Filter: An Application to Face Recognition*, Computer Vision and Image Processing, , Springer, IITDM, Jabbalpur, 2019

6. T. Ghosh, **Pradipta K. Banerjee**, A *Class Specific Representation Learning for Illumination Tolerant Face Recognition*, 2018, Recent Trends in Image Processing and Pattern Recognition, IAPR (endorsed), Solapur University, 2018

7. A. Amith, R.K.Saha, **Pradipta K. Banerjee**, *Unconstrained Band-pass Optimized Correlation Filter (UBoCF): An application to face recognition*, IEEE Applied Signal Processing Conference(ASPCON), Jadavpur University,December7-9, 2018

8. Papia Banerjee, **Pradipta K. Banerjee**, Asit K. Datta, *Face Detection and Landmark Localization using Non-parametric Approach and Landmark Filters*, *The Eighth International conference on Advances in Pattern Recognition, IEEE computational Intelligent Society*, Jan 4-7, 2015.

9. Papia Banerjee, **Pradipta K. Banerjee**, *Phase-Eigen Subspace based Illumination Invariant Face Recognition using Associative Memory*, *IEEE National Conference on Computing and Communication Systems*,2012, India

10. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Feature Based OT Parameter Selection of Correlation lter for Real Time Face Authentication in Frequency Domain*, *ICCCS, ACM, NIT Rourkella*, Odisha, India, 2011

11. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Face Detection and Eye Localization in Video by 3D Unconstrained Filter and Neural Network*, A. Abraham et al. (Eds.) : *ACC 2011, Part II , CCIS 191, pp.480 489, Springer-Verlag Berlin Heidelberg* , 2011, India

12. **Pradipta K. Banerjee**, Asit K. Datta, *An Unconstrained Frequency Domain Correlation Filter for Face Recognition with Reduced Misclassication Rate*, *Proceedings of International Conference on Trends in Optics and Photonics*, Dec 7-9, 2011, Kolkata, India

13. **Pradipta K. Banerjee**, *An approach of eye detection under varying in-plane rotation of face image, International Conference on Recent Trends in Information Systems, IEEE Computer Society*, 2011, India

14. Jayanta K. Chandra, **Pradipta K. Banerjee**, Asit K. Datta, *Singular value decomposition method for the detection of defect in woven fabric refined by morphological operation*, *IEEE Recent Advances in Intelligent Computational Systems*, 2011, India

15. Jayanta K. Chandra, **Pradipta K. Banerjee**, Asit K. Datta, *Modular eigen fabric based classifier for the detection of defect in woven fabric*, *IEEE Recent Trends in Information Systems*, 2011,India

16. Jayanta K. Chandra, **Pradipta K. Banerjee**, Asit K. Datta, *PCA and ANN based morphological process for the detection and classification in woven fabric*, *International Conference on Soft-computing and Engineering Applications*, 2011,India

17. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Wavelet Modied Phase-only Maximum Average Correlation Height Filter for Face Recognition*, *Proceedings of the Second International Conference on Intelligent Human Computer Interaction, Springer*, 2009, IIIT Allahabad,India

18. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *SVM Classier for Face Recognition Based on Unconstrained Correlation Filter*, *IEEE International Conference on Signal and Image Processing Applications*, 2009,Malaysia

19. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Illumination Invariant Face Recognition: Comparison of Performances of Fourier Domain Correlation Filters*, *Proceedings of International Conference on Trends in Optics and Photonics*, 2009, Kolkata, India

20. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *A Study of Frequency Domain Correlation Filter in Face Recognition*, *National Conference on Modern Trends in Electrical Engineering*, 2009,Hoogly, India

21. **Pradipta K. Banerjee**, Jayanta K. Chandra, Asit K. Datta, *Optical Implementation of Face Recognition System Based on Unconstrained Minimum Average Correlation Energy Filter*, *Proceeding of International Conference of Trends in Optics and Photonics*,December 13-17, 2008, IIT Delhi, India

22. Jayanta K. Chandra, **Pradipta K. Banerjee**, Asit K. Datta, *Morphological Reconstruction Operation for the Detection of Defects in Woven Fabric*, *IEEE Region 10 Colloquium and the Third ICIIS*, 2008

Research Interest

1. Broadly: interested in Machine Learning, Deep Neural Network, Image processing and Pattern recognition

2. Specifically: interested in Multimodal Emotion Analysis, and ATR

Salient Features in Research Experience

- Regression correlation filters
- Class specific representation learning
- Correlation filter(classifier) development, performance evaluation and modification towards pattern recognition and object detection
- Performance improvement of correlation filters with different image processing applications under different illumination conditions, expression variations and noisy conditions of facial images
- Nonlinear correlation filtering in class specific subspace approach

Research skills

1. Working platform : Python, Matlab, Latex

2. Algorithm developed : Machine Learning Algorithms, Deep Neural Net, Correlation filters, Regression Classifiers in Frequency Domain, Subspace methods, Representation learning, Clustering Algorithms

Accomplishments

1. Succesfully completed "**Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning**" course from Deeplearning.ai with 97.73% grade on Nov'2022

2. Succesfully completed "**Convolutional Neural Networks in TensorFlow**" course from Deeplearning.ai with 98.12% grade on Dec'2022

3. Succesfully completed "**Supervised Machine Learning: Regression and Classification**" course from Deeplearning.ai with 99.6% grade on Dec'2022

4. Succesfully completed "**Machine Learning**" course from Stanford University(COURSERA) with 93.7% grade on Feb'2016

5. Succesfully completed "**Neural Networks and Deep Learning**" course from deepleaning.ai (COURSERA) with 96% grade on 2017-2018

6. Succesfully completed "**Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization**" course from deepleaning.ai (COURSERA) with 95% grade on 2017-2018

Professional Experience

2005–till date	**Future Institute of Technology, Kolkata, Future Institute of Engineering and Management, Kolkata** Professor & Head, Department of CSE-AIML(Jan' 2022- till date) Professor & Head, Department of Electrical Engineering(July' 2005-Dec' 2021) *Courses taken (Theoretical)* : Machine Learning, Deep Learning, Pattern recognition, Computer Vision *Courses taken (Practical)* : Machine Learning, Deep Learning)

Teaching Experience

19+ years

Professional Development Activities

Funded Project

- **Position** : Principal Investigator
  Title of Project : Automatic Traffic Detection and Counting on Large Scale Real-Time Database on various road types in Kolkata and Howrah Municipal Corporation.
  Organisation : WBPCB and The Energy and Resources Institute, Delhi
  Year : 2023-24
  Amount : 4.5 Lacs
  Status : Completed

Project Developed (National Recognition)

- **Position** : Mentor/Team guide
  Title of Project : A small scale quadcopter proto for real time activity detection and classification
  Responsibilities : Developed state-of-the-art detection and classification algorithm using deep neural network in Python
  Organisation : Defense Research Unmanned System Exposition (DRUSE) organized by

DRDO
Year : 2018
Status : Completed (Finalist)

- **Position** : Mentor/ Team guide
  Title of Project : A small-scale Quadcopter proto for real time activity detection, classification and localization using deepnet and hybrid correlation filtering
  Responsibilities : Developed state-of-the-art detection and classification and localization algorithm
  Organisation : AERO INDIA STUDENT PAVILION, Bangalore
  Year : 2019
  Status : Bagged 6th position

- **Position** : Mentor/Team guide
  Title of Project : Automatic Smart Dustbin : A neural network and image processing application, ***Swachh Bharat Abhiyan Project***
  Resposibilities : Developed Convolutional Neural Network(CNN) to classify bio-degradable, bio-medical and no-biodegradable objects using image processing technique in Python
  Organisation : INAE Youth Conclave organized by IIT KGP
  Year : 2018
  Status : Bagged First Position

Ph.D Enrollment

Position:	Principal Supervisor
Ph.D Scholar:	Debosmita Chakraborty
Date of Enrollment:	05.10.2020
Title:	Some Studies on Multimodal Deep Learning Techniques and Frequency Domain Analysis for Enhanced Human Emotion Recognition and Mental Health Assessnent.
University:	MAKAUT, West Bengal
Progress :	Scopus indexed Journal (1), Conference (1), Communicated (Conference) : 1

Laboratory Development (Interdisciplinary)

1. **Developed IOT (Internet of Things) project laboratory, based on Espressif and Raspberry Pi systems**.
   (a) Smart home automation system through Mobile app development.
   (b) Controlling devices through Firebase and MQTT protocol
   (c) Both Local and Remote device controlling through Wifi, etc.
2. **Developed Machine Learning and Computer Vision Laboratory** .
   (a) Both CPU and GPU applications through Google cloud
   (b) Object tracking system with activity classifications

Membership

- Association for Computing Machinery (**ACM**)
- Computer Society of India **(CSI)**
- International Association of Computer Science and Information Technology (**IACSIT**)
- International Association of Engineers (**IAENG**)

Reviewer

- International Journal of Pattern Recognition and Artificial Intelligence (IJPRAI)
- IEEE Signal Processing Letters
- Engineering Applications of Artificial Intelligence, Elsevier Press
- International Conference on Science, Engineering and Technology
- Computer Vision and Image Understanding, Elsevier Press
- IEEE conference on Recent Trends in Information Systems

Workshops/Seminar/Conference Attended/Organized

24th – 25th March 2019	**Convener** : Organized a two day seminar on Computational Intelligence in Signal Power and Control (CISPC19), F.I.E.M. Kolkata
22nd Mar 2018 – 23rd Mar 2018	**Convener** : Organized a two day seminar on "Computational Intelligence in Signal Power and Control (CISPC'18), F.I.E.M. Kolkata
23th Mar 2017 – 24th Mar 2017	**Convener** : Organized a two day seminar on "Computational Intelligence in Engineering Applications (CIEA'17), F.I.E.M. Kolkata
24th Jun 2013 – 5th Jul 2013	Training Programme on Digital Signal Processing Speech & Image Processing Organized by C-DAC, Kolkata as a part of Advanced Faculty Training Programme, HRD Division, Govt. Of India for a period of two weeks
23rd Feb 2013	National Seminar on SMART Metering under UGC-SAP DRS-I project, Department of Applied Physics, University of Calcutta, Kolkata
30th Jan 2012 – 10th Feb 2012	AICTE sponsored Staff Development Program on Optimization of Decision Tools using Soft Computing Models & MATLAB, at Kalyani Govt. Engg. College

Honours and Awards

1999	Bela Rani Medal, proficiency in B.Sc. Part-II (Physics Hons.)
1999	Sudhir Kumar Bhose Memorial Medal, proficiency in B.Sc. Part-II (Hons.)
1999	Niladri Bhattacharyay Memorial Award, proficiency in B.Sc. Part-II (Physics Hons.)
1999	Manada Sundari Stipend, proficiency in B.Sc. Part-I (Physics Hons.)
1998	H.L.Mitter Memorial Prize, proficiency in B.Sc. Part-II (Hons.), 1999

Summary

Currently working as a Professor & Head in the Dept. of CSE-AIML, at F.I.T., Kolkata, with 19 years of teaching experience. My research interest includes machine learning, neural network, pattern analysis. I have published One (1) Book, Five (5) Book Chapters, Seven(7) SCI Journals, Two (2) Scopus indexed journals, and Twenty One (21) International Conferences. For quick reference about my academic performance and research works please visit my web page https://sites.google.com/view/pradiptakbanerjee

I hereby declare that the details and information given above are complete and true to the best of my knowledge



Signature